
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Geometric Corrections

Geometric Correction

- It is usually necessary to preprocess remotely sensed data and remove geometric distortion so that individual pixels are at their proper planimetric (x, y) map locations
- This allows remote sensing derived information to be related to other thematic information in GIS or spatial decision support systems (SDSS)
- Geometrically corrected imagery can be used to extract accurate distances, polygon area, and direction (bearing) information

Geometric Correction

- **Geometric correction of remotely-sensed images is required when these images, or a product derived from it are to be used:**
  - **To transform an image to match a map projection**
  - **To locate points of interest on map and image**
  - **To overlay temporal sequences of images of the same area, acquired by different sensors, and**
  - **To overlay images and maps for combined GIS data analysis**

Geometric Errors

- **Remotely sensed imagery may have**
  - **Internal geometric errors**
  - **External geometric errors**
- **It is important to recognize**
  - **The source of the internal and external errors and**
  - **whether it is *systematic* (predictable) or *nonsystematic* (random)**
    - **Systematic geometric error is generally easier to identify and correct than random geometric error**

Internal and External Geometric Errors

- **Internal geometric error (*systematic)***
  - **Introduced by the remote sensing system itself or in combination with Earth rotation or curvature characteristics**
- **External geometric error (Un- *systematic)***
  - **Usually introduced by phenomena that vary in nature through space and time. The most important external variables that can cause geometric error in remote sensor data are random movements of the platform at the time of data collection**

Geometric Errors

- **Sources of errors**
  - **Systematic**
    - **Earth's rotation- effect is to skew the image**
    - **Relief distortion**
    - **Panoramic distortion: a scale distortion (an increase in the resolution cell size) for observations away from nadir due to an increase in the area viewed by the sensor.**
    - **Earth curvature**

Geometric Errors

- **Sources of Errors**
  - **Unsystematic (Random)**
    - **Attitude of sensor (Roll, Yaw, Pitch)**
    - **Altitude of sensor (Variations in altitude)**
- **All sources of error contribute unequally to the overall geometric distortion in an image**

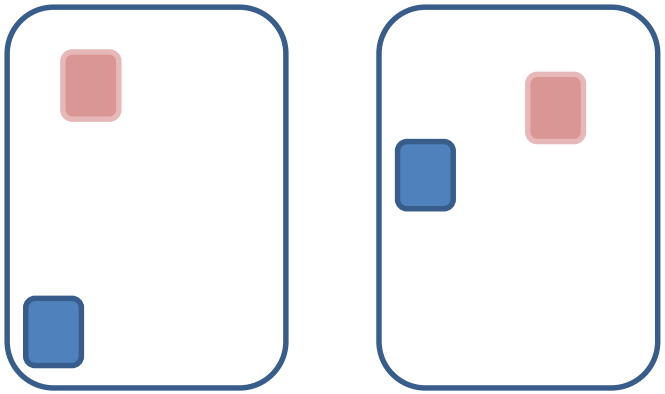
Systematic Errors Correction

- **Errors that can be corrected using data from platform ephemeris and knowledge of internal sensor distortions**
- **Mostly removed after data acquisition (by Satellite data providers)**
- **Easily corrected by formulas that model the distortions**
  - **(Ephemeris:** calculated positions, velocities etc of a celestial object at regular intervals throughout a period)

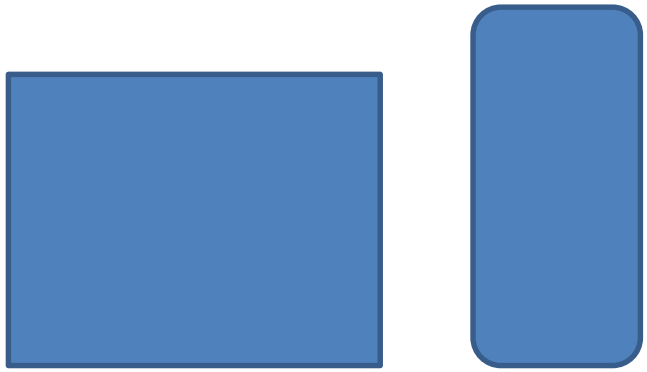
Un-Systematic Errors Correction

- **Errors that can not be corrected with acceptable accuracy without a sufficient number of GCPs**
- **May remain in image, making it non-planimetric (not in proper x,y locations)**
- **For Correction, a sufficient number of GCPs are required**

Distortions



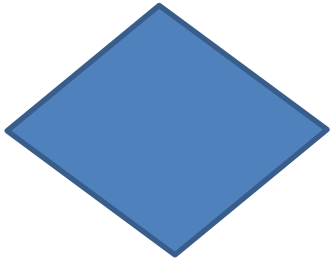
Translation in X and Y



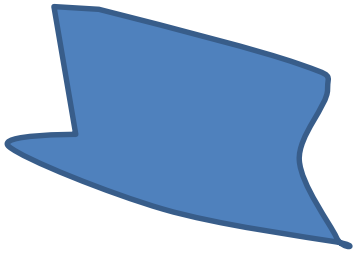
Scale change in X and Y



Skew



Rotation



Warp (bend)

Ground Control Points (GCP)

- **A GCP is a location on the surface of the Earth (e.g., a road intersection) that can be identified on the imagery and located accurately on a map**
- **A point on the surface of the earth where both image coordinates (row, column) and map coordinates(Lat/Lon, feet, meters) can be identified**

Ground Control Points

- **Sources to obtaining accurate *ground control point* (GCP) :**
  - **Hard-copy planimetric maps**
  - **Digital planimetric maps**
  - **Global Positioning System (GPS) instruments**
  - **Field Surveys**
  - **Digital orthophotoquads (DOQs)**
    - **that are already geometrically rectified using "Differential Rectification" for tilt and terrain relief (e.g., U.S.G.S. digital orthophoto quarter quadrangles —DOQQs)**
  - **("Differential Rectification" :**Process same as standard rectification, but to small patches to maintain common scale)
  - **Planimetric"** refers to the representation or measurement of features on a two-dimensional, flat plane, independent of elevation or relief.

Image Registration/Geo-referencing

Correction of Geometric Distortion

- There are two techniques that can be used to correct the various types of geometric distortion
- One is to model the nature and magnitude of the sources of distortion and use these models to establish correction methods.
- This technique is effective when the types of distortion are well characterized, such as that caused by earth rotation.
- The second approach depends upon establishing mathematical relationships between the addresses of pixels in an image and the corresponding coordinates of those points on the ground (via a map)

Ground Control Points

- **Geometric distortions introduced by sensor system *attitude* and/or *altitude* changes, can be corrected using ground control points (GCP) and appropriate mathematical models**

Ground Control Points

- **Image analyst must be able to obtain two distinct sets of coordinates associated with each GCP:**
  - ***Image coordinates* specified in *i* rows and *j* columns, and**
  - ***Map coordinates* (e.g., *x, y* measured in degrees of latitude and longitude, feet in a State Plane coordinate system, or meters in a Universal Transverse Mercator projection)**
- **The paired coordinates (*i, j* and *x, y*) from many GCPs can be modeled to derive *geometric transformation coefficients***
- **These coefficients may be used to geometrically rectify the remote sensor data to a standard datum and map projection**

Geometric Corrections

- **Purpose:**
  - Compensate for spatial distortions, to rectify (align) an image with reference to a map or an image
    - **Rectification:**
      - The process of spatially fitting image to map projection
      - ***Image-to-map rectification or image-to-image rectification***
    - **Registration:**
      - The process of spatially fitting image to another image or
      - Fitting of the coordinate system of one image to another image of the same area
      - ***Image-to-image registration***

Types of Geometric Correction

- **Commercial remote sensor data already have much of the *systematic error* removed**
- **Unless otherwise processed, *unsystematic random error* remains in the image, making it non-planimetric**
- **The *general rule of thumb* is to rectify remotely sensed data to a standard map projection whereby it may be used in conjunction with other spatial information in a GIS to solve problems**

Image to Map Rectification

- **Process to make the geometry of an image planimetric (i.e.to be in proper x,y position)**
- **Used for the measure of accurate area, direction and distances**
- **May not remove all distortions (topographic relief)**
- **GCPs used to fit image to the map geometry**

Image to Image Registration

- **Process by which two images of like geometry and of same geographic area are positioned coincident with respect to one another**
- **Used when it is not necessary to assign unique x,y coordinates to each pixel**
- **May be used to simply compare two images acquired at different times for change detection**

Image to Image Registration

- **Mostly the image is rectified to some reference system**
- **Same principle are used for both rectification and registration**
- **In rectification, reference is standard map and in registration reference is another image**
- **Both methods rely on human interaction**

Levels of Geometric Rectification

- **Level 1 (zero ,first level)**
  - **Mostly Systematic Errors corrected (original rectification)**
- **Level 2**
  - **Use GCPs (X, Y) and transformation to rectify image (Geometric rectification), when area is less and flat**
- **Level 3**
  - **Use 3D GCPs, (DEM) and transformation (ortho rectification)**

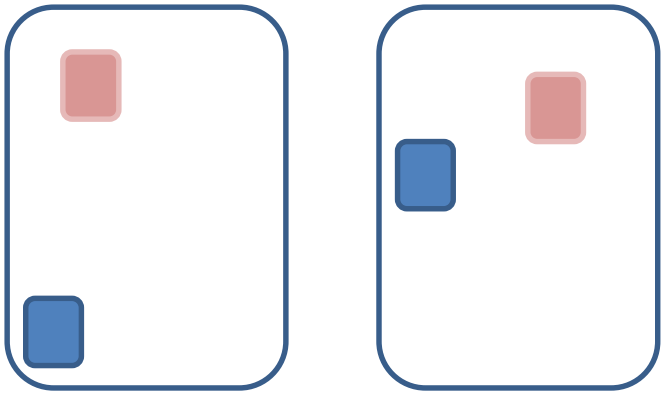
GEOMETRIC TRANSFORMATION/ CORRECTION

Image Registration/Geo-referencing

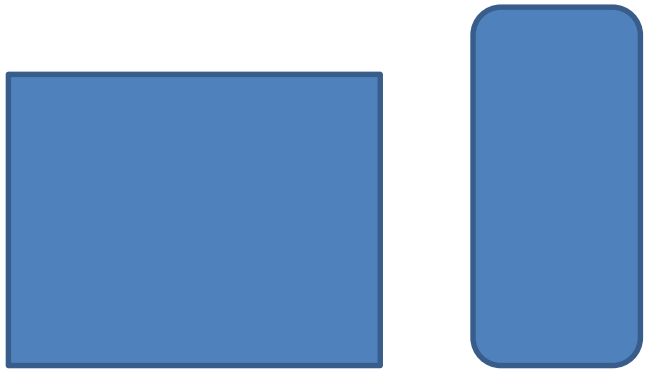
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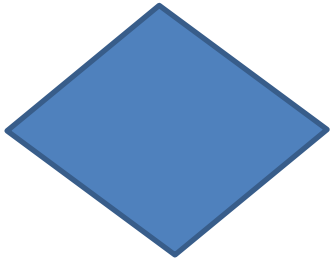
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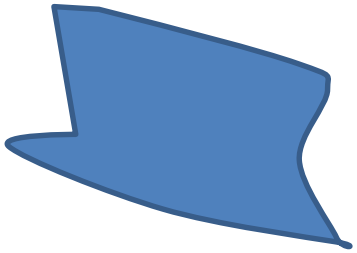
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Geometric Correction

- **Process includes**
  - ***Spatial interpolation***
    - Determination of a relationship between the coordinate system of the image and map or image and image
  - ***Intensity interpolation***
    - To assign proper input BV to output pixel locations

Spatial Interpolation

- **The geometric relationship between the input pixel coordinates (column and row; referred to as x', y') and the associated map coordinates of this same point (X, Y) must be identified**
- **Number of GCP pairs are used to establish the nature of the geometric coordinate transformation**

Intensity (Spectral) Interpolation

- A pixel in the rectified output image often requires a value from the input pixel grid that does not fall neatly on a row-and-column coordinate
- Unfortunately, there is no direct one-to-one relationship between the movement of input pixel values to output pixel locations
- When this occurs, there must be some mechanism for determining the brightness value (*BV*) to be assigned to the output rectified pixel

Geometric Rectification via GCPs

- Corresponding points are selected (usually manually) from map/image or reference images
- Transformation is computed to transform distorted image to map/reference image:
  - x = f1(X,Y)
  - y = f2(X,Y)
  - where:(x,y) = distorted-image coordinates (col, row)
  - (X,Y) = correct (map) coordinates
  - (f1,f2) = transformation functions created using ground control points (GCP's)
- Commonly low order polynomials are used to accomplish transformation

Polynomial Models

- To relate two-dimensional ground coordinates to image coordinates
- Equation for 1st order polynomial, two equations for each GCP
  - $x = a0 + a1X + a2Y$
  - $y = b0 + b1X + b2Y$
- This transformation accounts for six parameters to establish this relation,
  - Shift(Translation) in x and y, Scale in x and y, Shear(skew), Rotation
- Minimum number of GCP's needed is 3
- For three GCPs, six equations can be written, and solved simultaneously for unknowns

Affine/ Polynomial Model

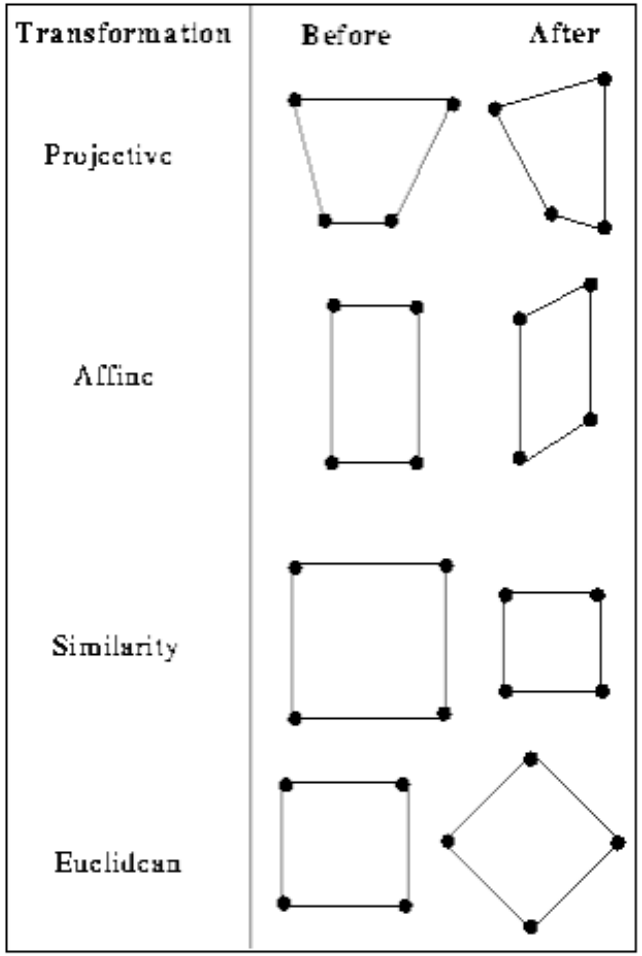
- **Equation for minimum number of GCPs**

  - **No of GCPs =** $\frac{(T+1)*(T+1)}{2}$

  Where T=order of transformation (polynomial)

- **For moderate distortions and relatively small areas, simple transformation is ok,**
- **For large area and more distortion, complex transformations are involved**

Transformations



Projective (where a square can be transformed into any more general quadrilateral
maps lines to lines (but does not necessarily preserve parallelism

preserves lines and parallelism (maps parallel lines to parallel lines)

shape-preserving transformations
proportional scaling transformation

Euclidean (where only rotations and translations are allowed)

Important Notes-GCPs

- **Use many more than minimum GCPs desirable**
- **Results from small number of GCPs should be viewed cautiously**
- **Spatial distribution of points also critical- to avoid biased fit,**

  Problem in sea, water areas, featureless regions(deserts)
- **Avoid GCPs from maps not revised for long period of time**

Image to Map Rectification

Selecting Ground Control Points for Image-to-Map Rectification

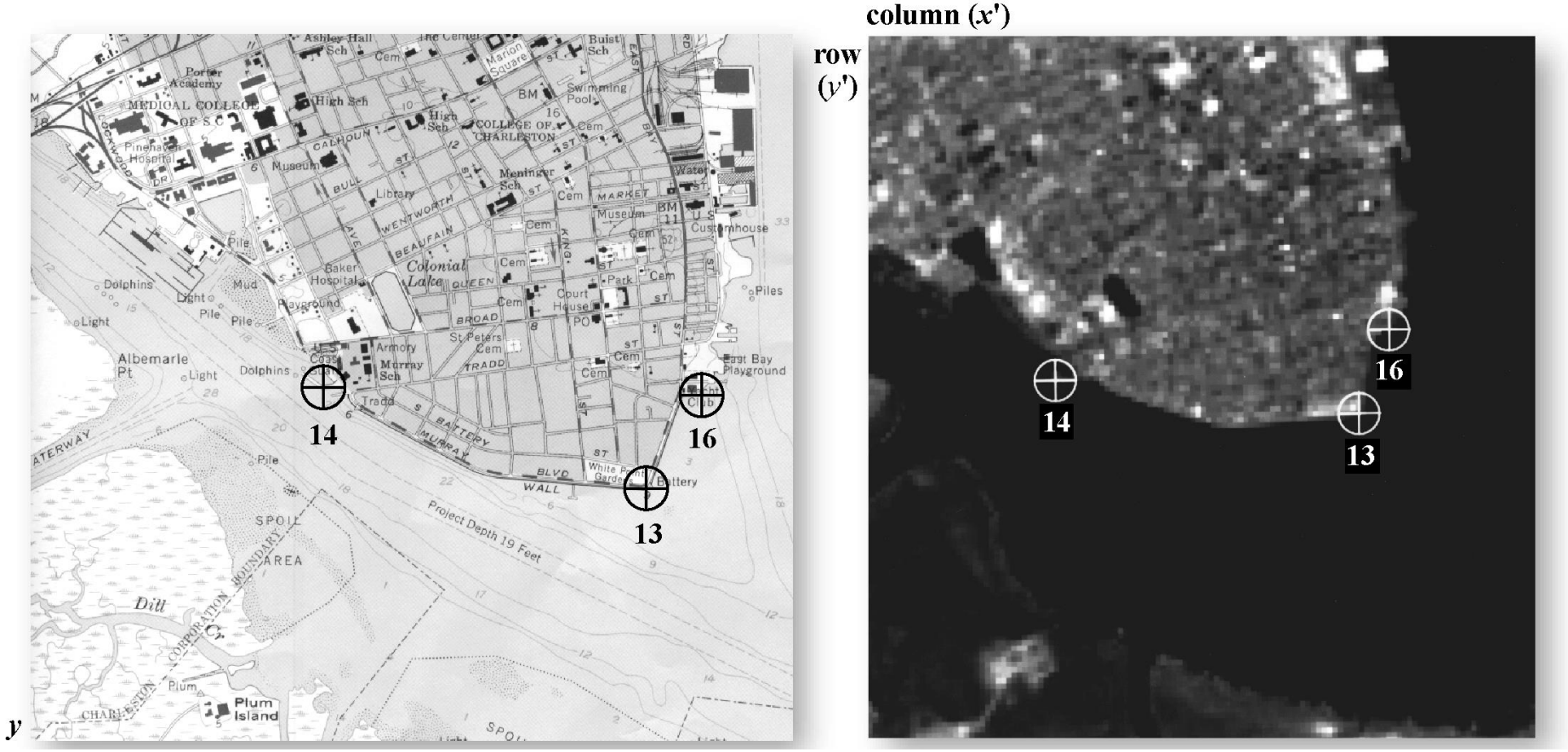
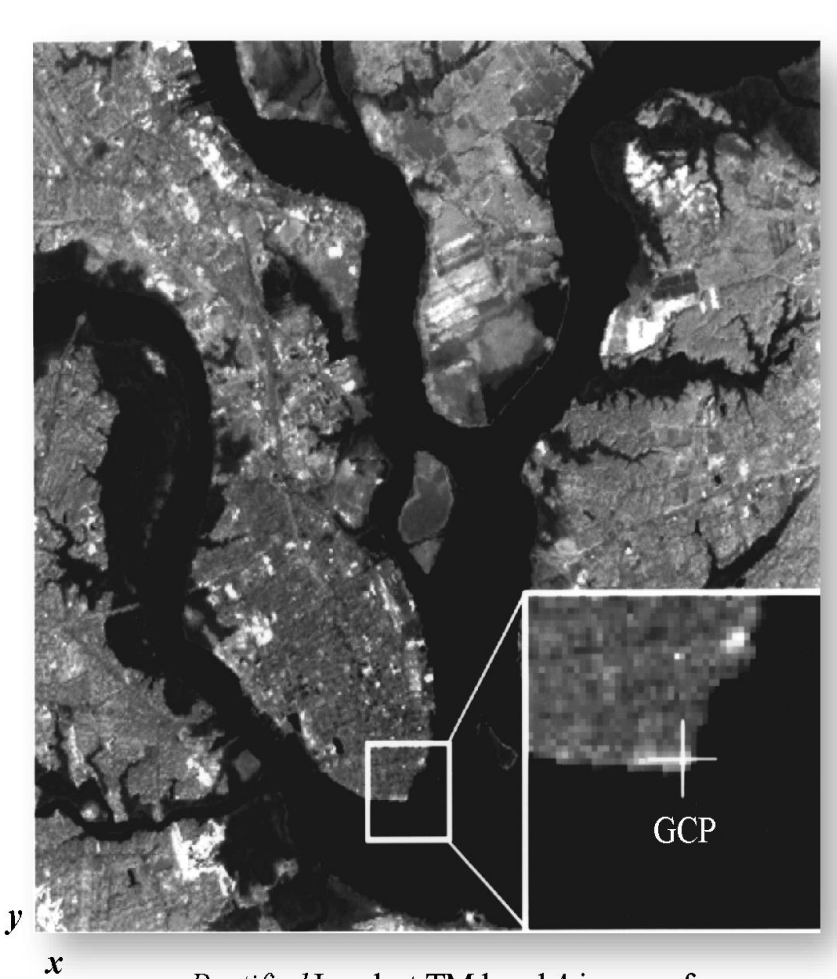
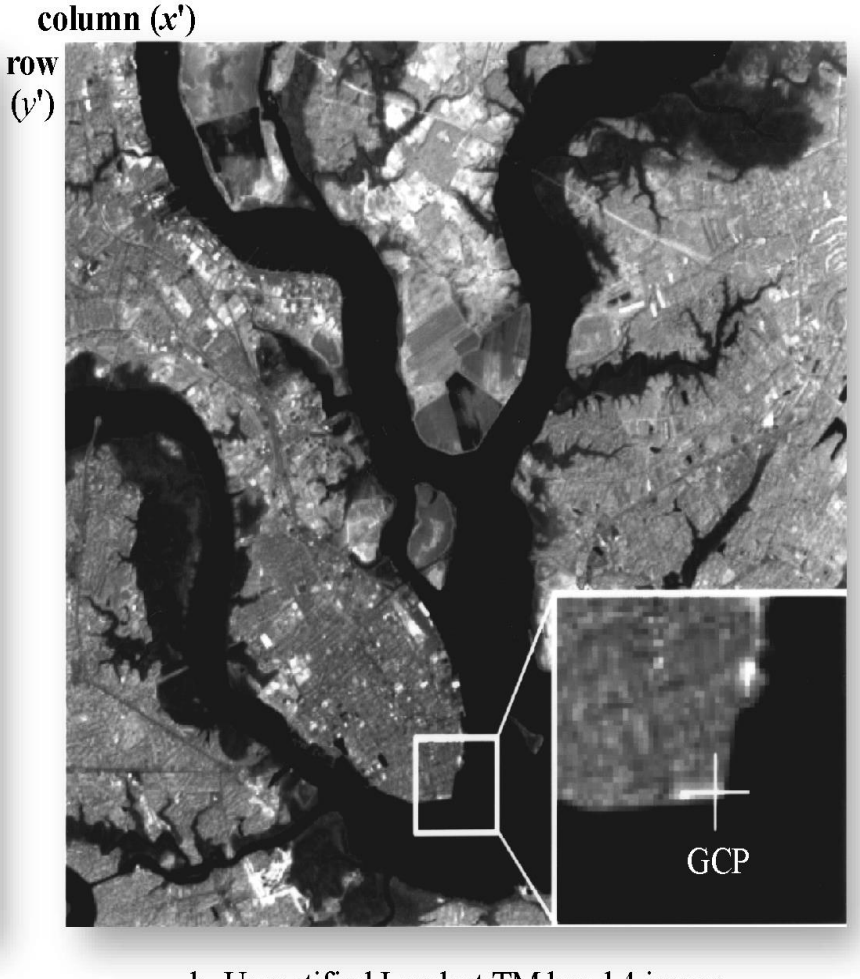


Image to Image Registration



a. *Rectified* Landsat TM band 4 image of Charleston, SC, obtained on November 9, 1982.



b. Unrectified Landsat TM band 4 image obtained on October 14, 1987.

Intensity Interpolation/ Resampling

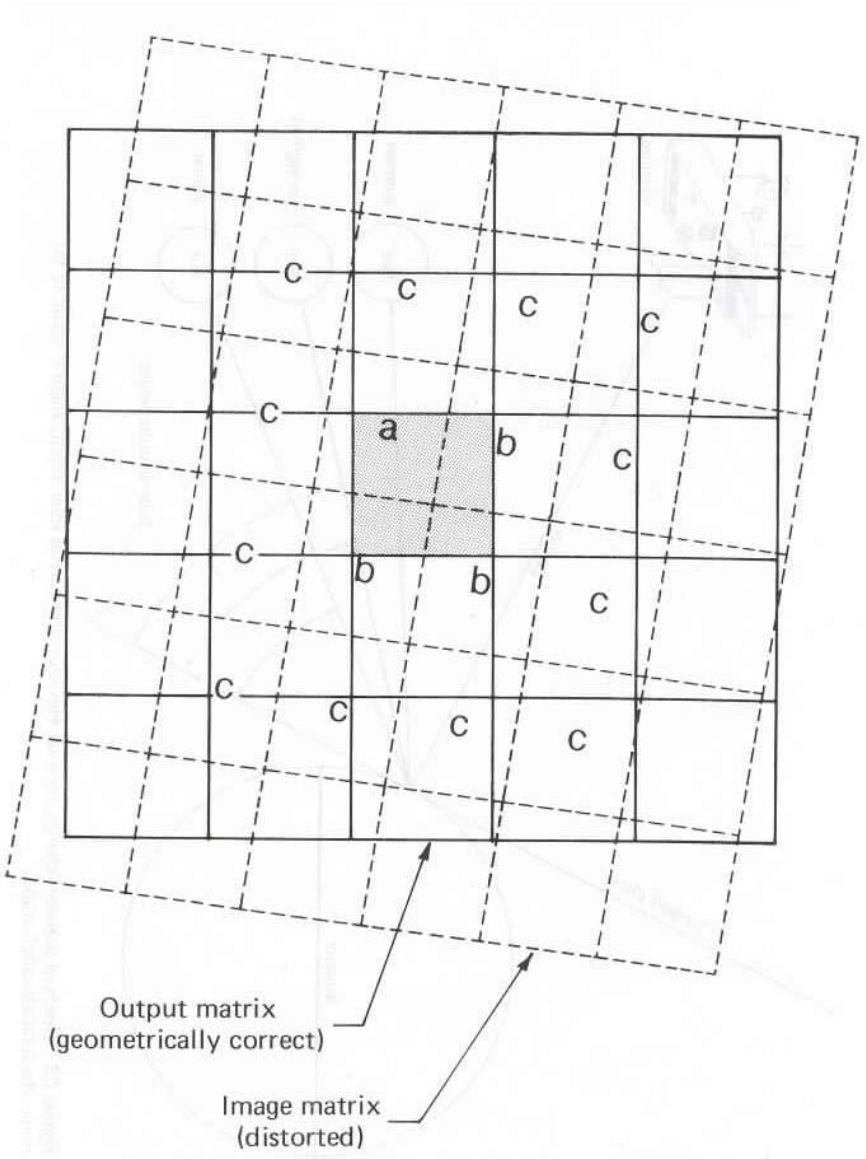
- **Interpolation/ resampling Methods**
  - **Nearest Neighbor (Zero order interpolation)**
  - **Bilinear interpolation (1st order interpolation)**
  - **Cubic convolution (Bicubic interpolation or 2nd order interpolation)**
- No one re-sampling method is optimal for all situations, depends upon the data application and the advantages and disadvantages of each

Interpolations

- **Nearest Neighbor**
  - **Assigns each corrected pixel, the value from the nearest uncorrected pixel**
  - **Maintains actual pixel values but results over boundaries are not smooth, Jagged edges**
  - **Recommended if data are to be further used for classification**
- **Bilinear**
  - **Calculates a value for each corrected pixel based on the weighted average of the four nearest input pixels, Result is interpolated pixel values(altered values)**
  - **Image has more smooth**
  - **Not used prior to classification, rather post classification**
- **Cubic convolution (Bicubic)**
  - **Calculates a value for each corrected pixel based on the weighted average of the four nearest input pixels, Result is interpolated pixel values(altered values)**
  -

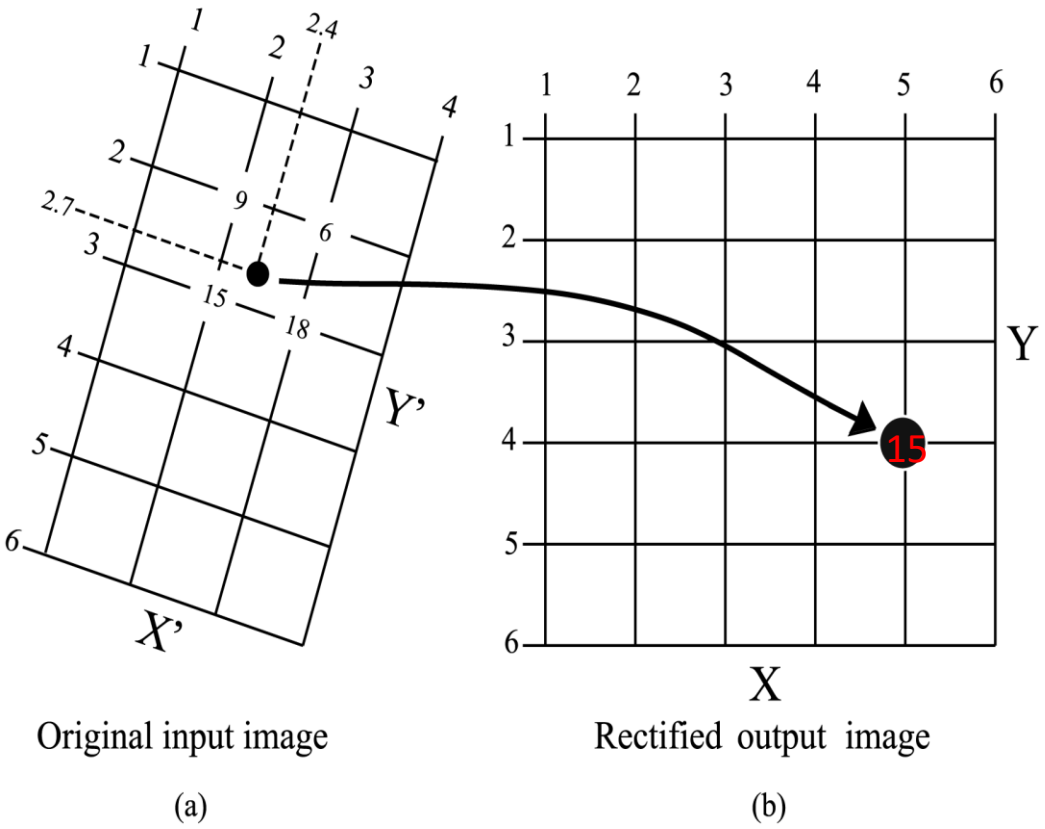
Intensity Interpolation

- **Nearest Neighbor**
- Select actual pixel whose center is closest to point located on the image
- Maintains actual pixel values, but results over boundaries are not smooth, Jagged edges
- Recommended if data are to be latter classified



Nearest-neighbor Resampling

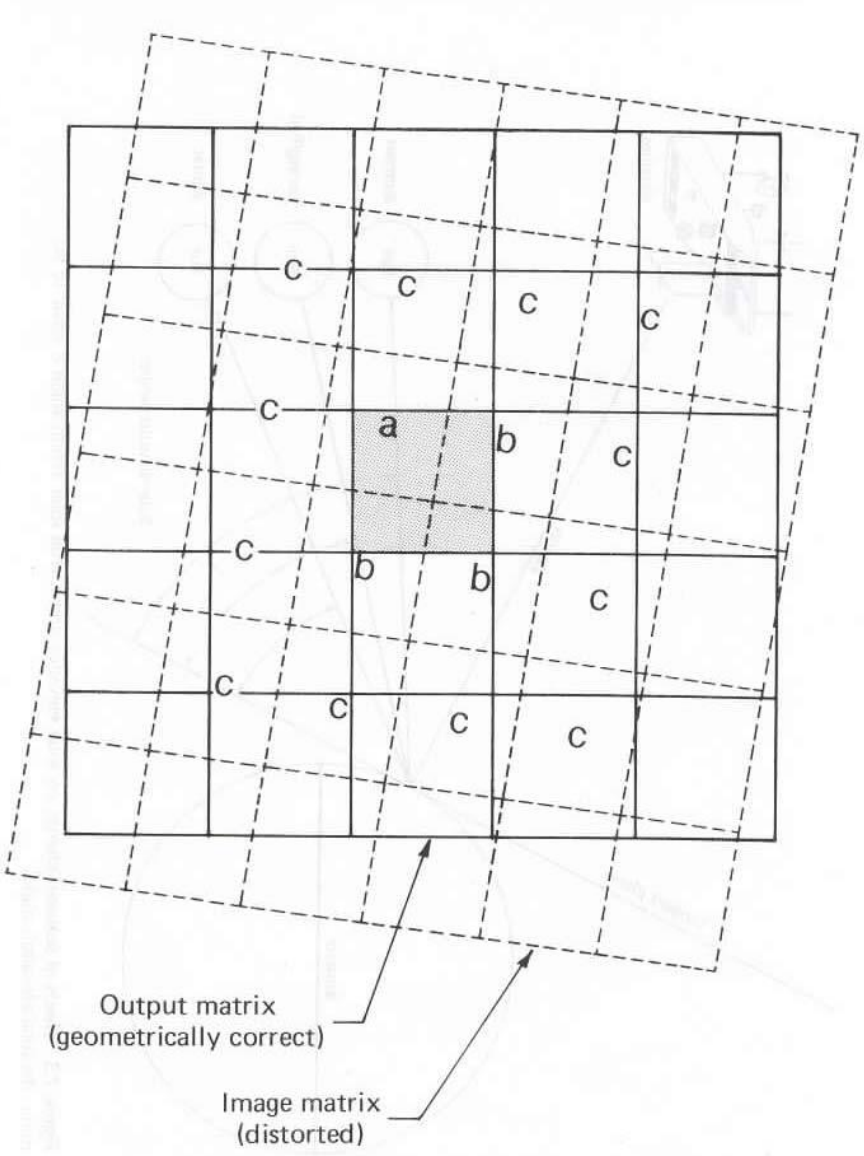
The brightness value closest to the predicted *x'*, *y'* coordinate is assigned to the output *x*, *y* coordinate



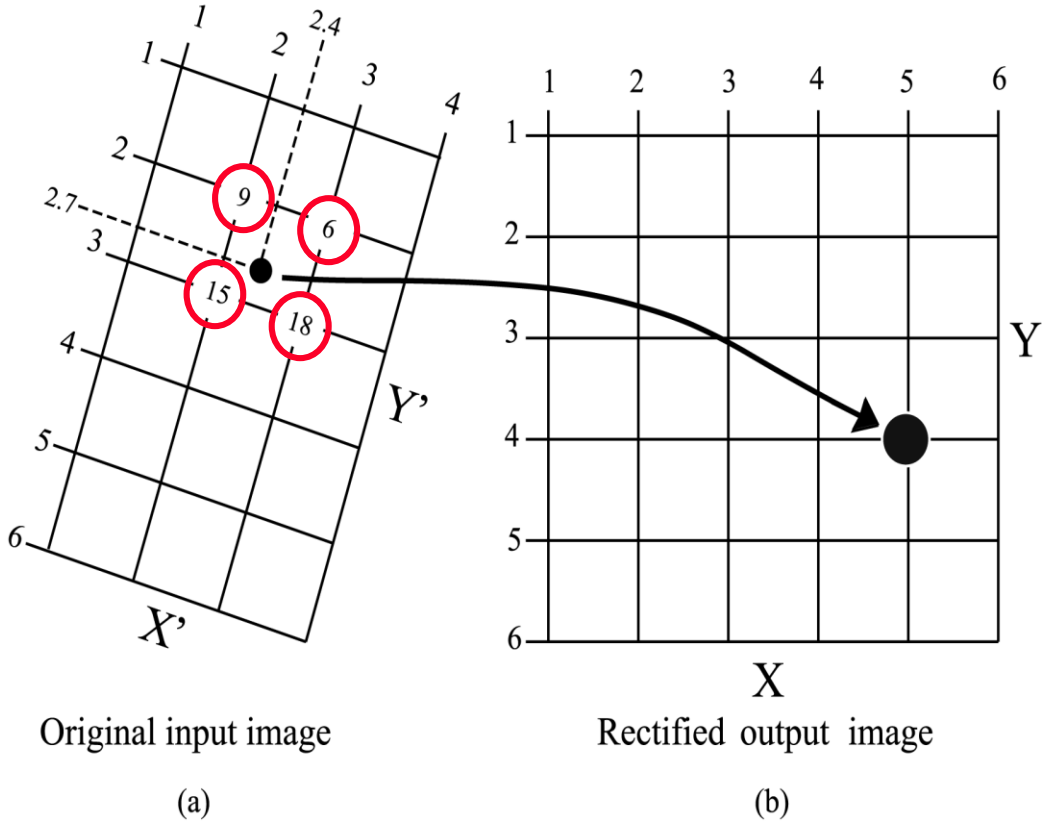
Intensity Interpolation

Bilinear interpolation

- Calculates a value for each corrected pixel based on the weighted average of the four nearest input pixels
- Image has more natural look, geometrically accurate
- Result is interpolated pixel values(altered values)
- Not used prior to classification, rather post classification



Bilinear Interpolation

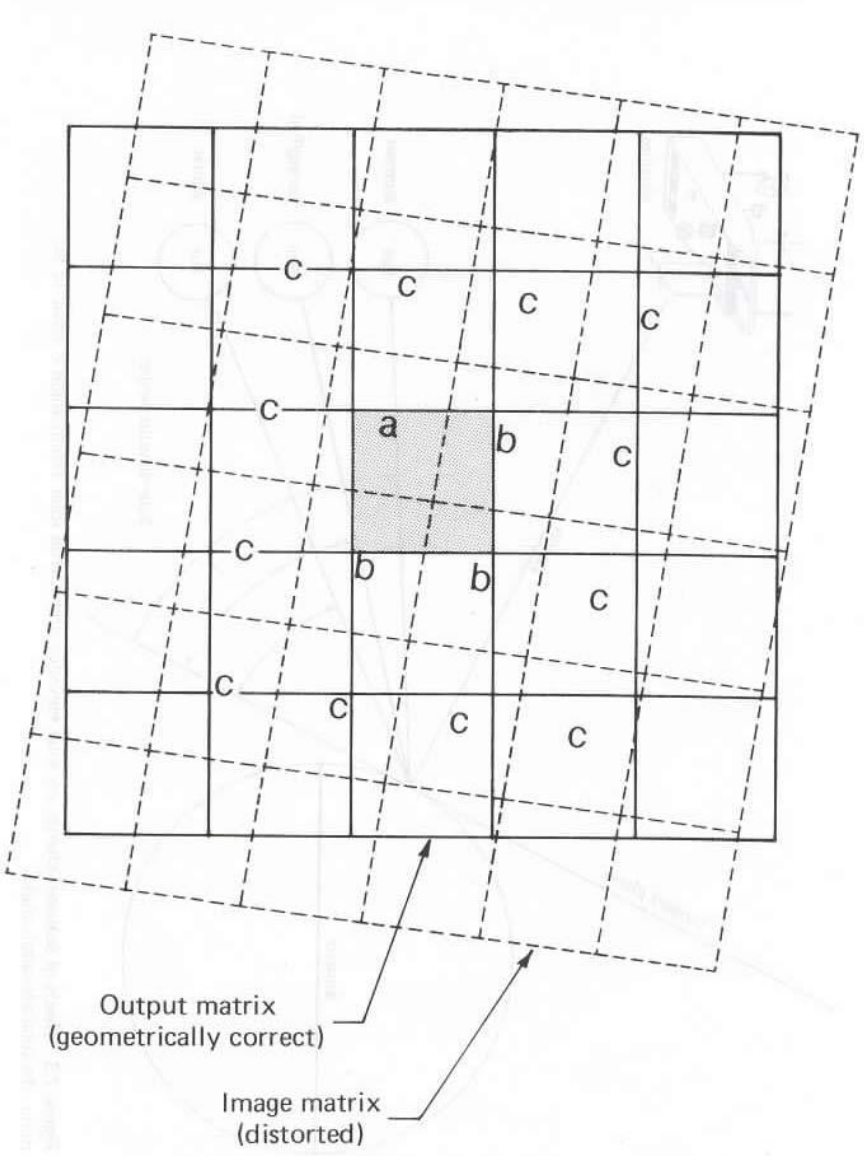


Interpolating brightness values of the ***4*** nearest pixels, then computes a new brightness value based on the weighted distances to these points to the desired position (***x', y'***). Closer a pixel is to the desired (***x', y'***) location, the more weight it will have in the final computation of the average.

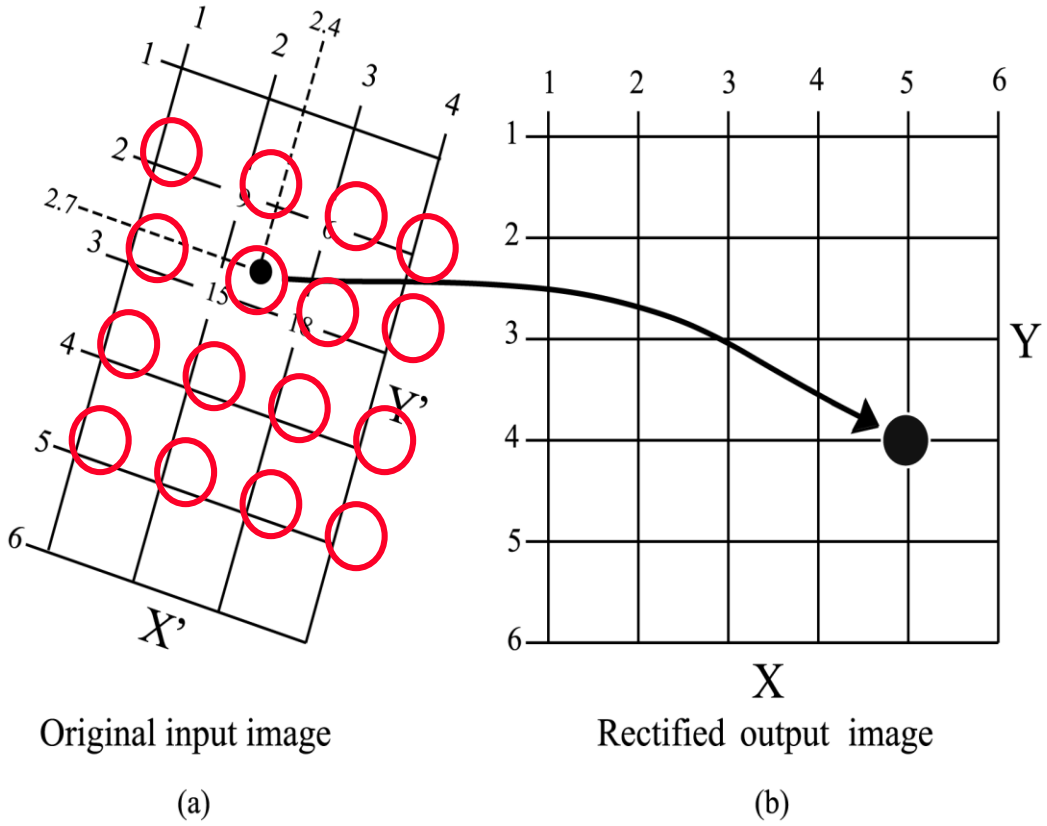
Intensity Interpolation

Cubic convolution

- Uses cubic polynomials and 16 surrounding pixels
- Result is interpolated pixel values, data values altered more than other methods
- Yields smooth and much more attractive appearance
- Computationally more intensive

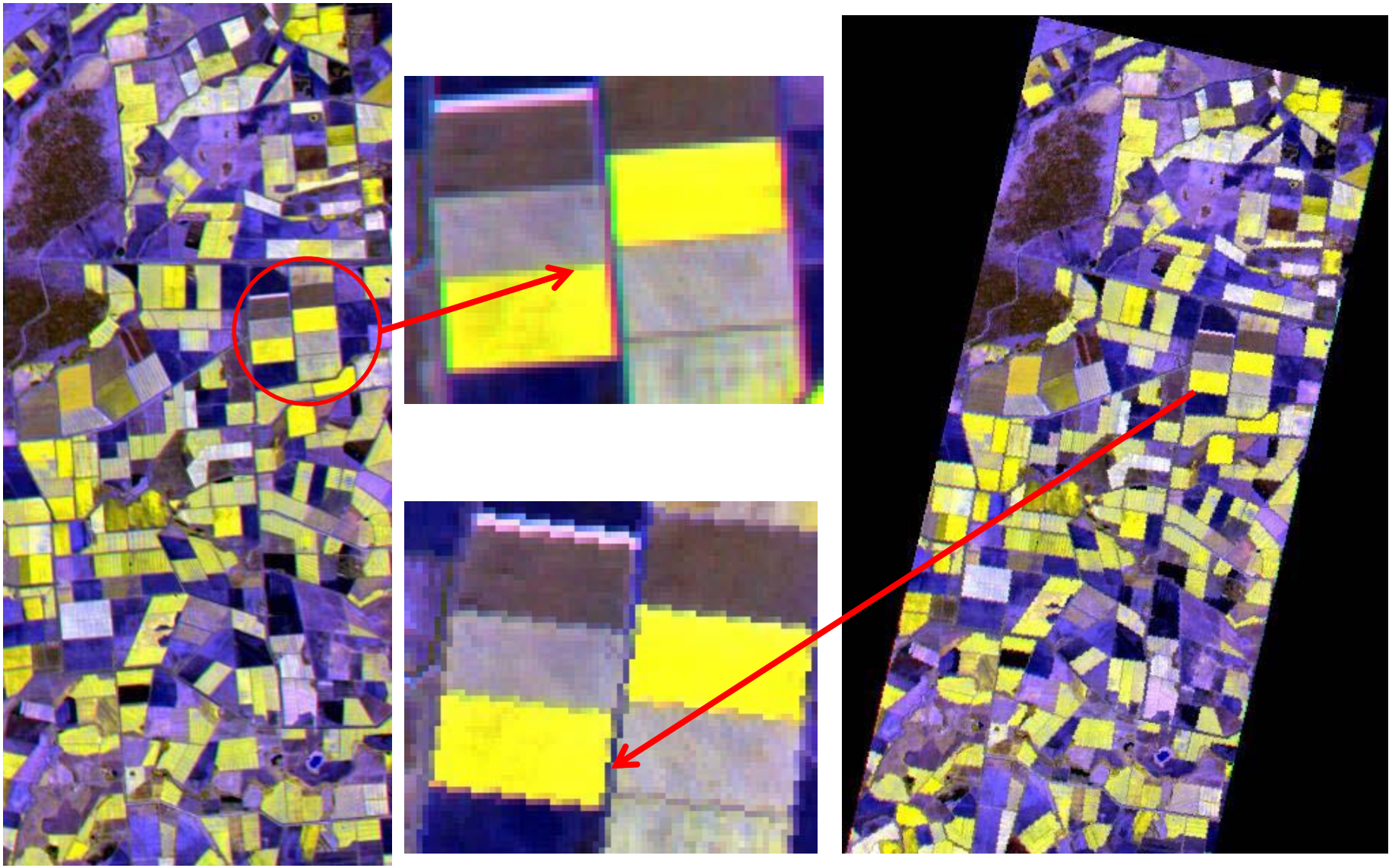


Cubic Convolution



Assigns values to output pixels in much the same manner as bilinear interpolation, except that the weighted values of ***16*** pixels surrounding the location of the desired pixel are used to determine the value of the output pixel.

Geo-Correction: Impact of Resampling



Error and Accuracy

- **The errors that remain after transformation, are called residual errors**
- **The magnitude of residual errors is an indicator of the quality of transformation**
- **Overall accuracy of transformation is usually expressed by the Root Mean Square Error**
- **RMS error calculates a mean value from the individual residuals**

RMS Error

- Helps to check which GCP contributes more error
- To find overall RMS error of the process
- If it does not meet the user defined threshold, then
  - Delete the GCP contributing higher error
  - Re-compute Transformation parameters
  - Re-compute Error

RMS Error

- **RMS error is calculated using the following equation:**

$$\text{RMS error} = \sqrt{(x_r - x_i)^2 + (y_r - y_i)^2}$$

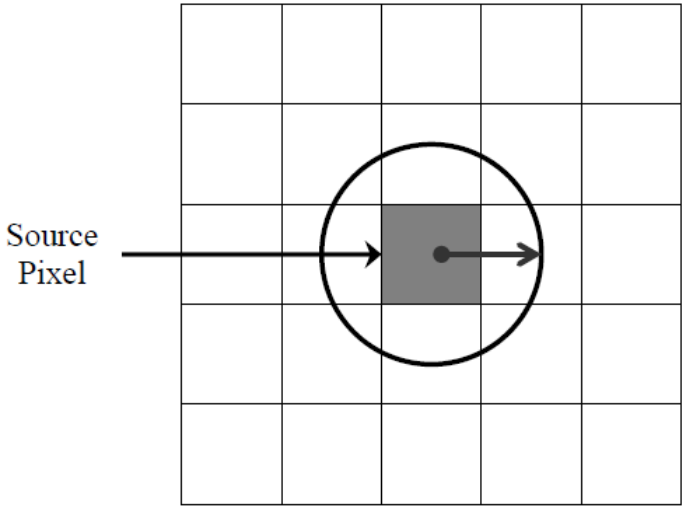
where:

- **x_i and y_i are the source coordinates in the reference image/ map**
- **x_r and y_r are the transformed coordinates**

RMS Error Tolerance

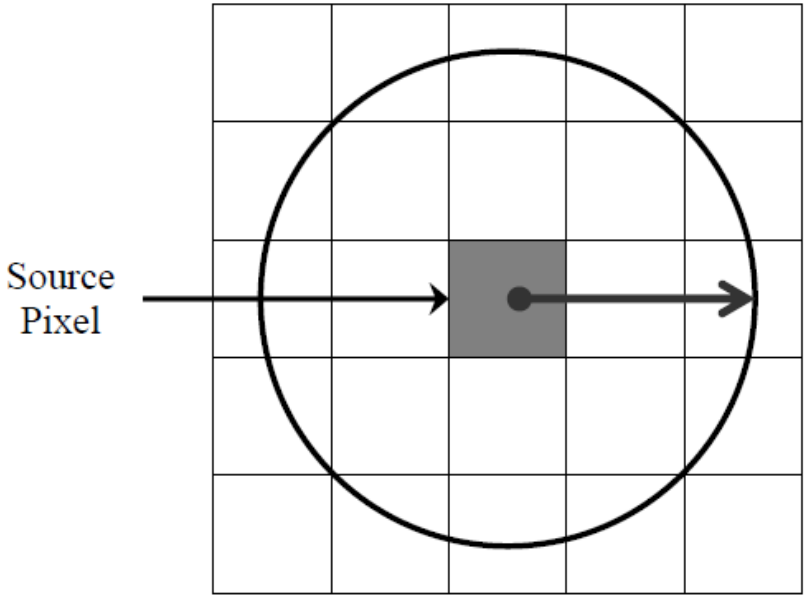
- **The RMS error test sets a window of pixels that is considered as correct**
- **For example, if the RMS error were to be set at 1.0, the output location would be considered correct if it falls within 1 pixel of the source**
- ***This means that the output location can be off by one pixel in either the x or y direction***

RMS Error Tolerance



RMS Error Tolerance = 1 pixel
(circle radius from center of Source Pixel)

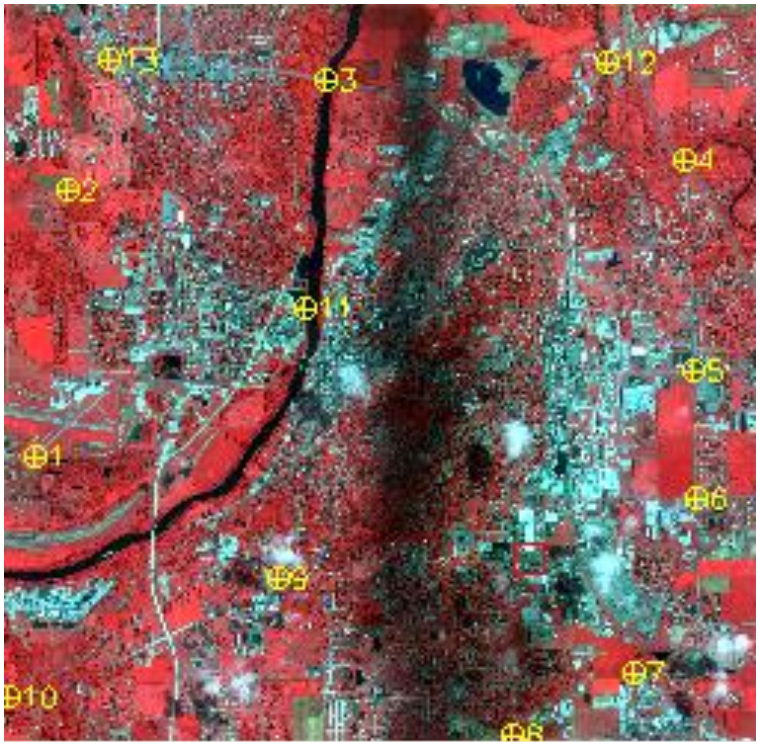
Transformed coordinates within the circle
are considered acceptable



RMS Error Tolerance = 2 pixels
(circle radius from center of Source Pixel)

Transformed coordinates within the circle
are considered acceptable

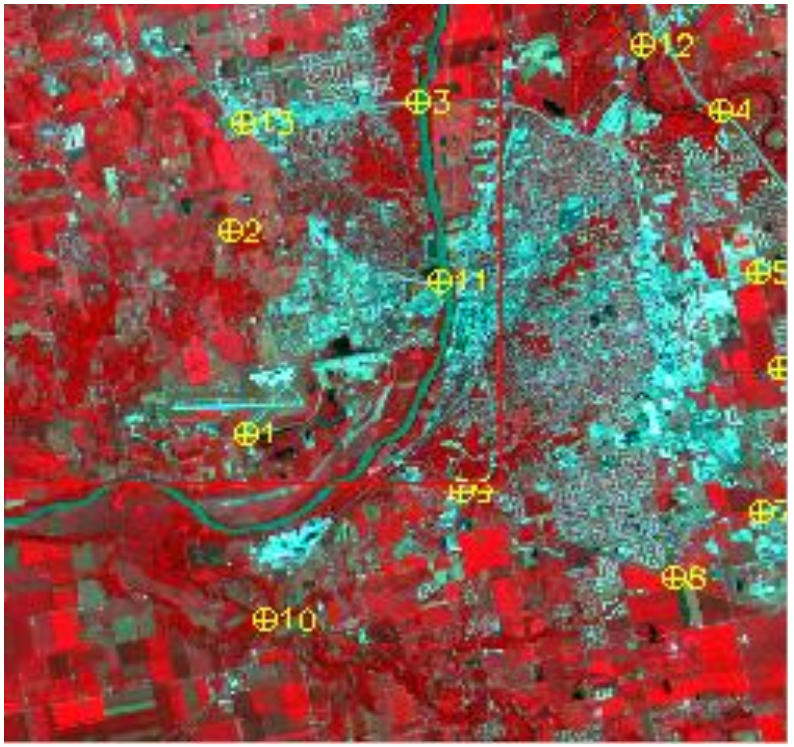
Image to Image Geo-referening



File Information:

File: F:\Image data store\Campus_000718\ik
Dims: 9784 x 9516 x 3 [BIP]
Size: [Unsigned Int] 558,628,082 bytes.
File Type : TIFF .
Sensor Type: Unknown
Byte Order : Network (IEEE)
Projection : UTM, Zone 16 North
Datum : North America 1983
X Pixel : 1.00000 Meters
Y Pixel : 1.00000 Meters
Wavelength : None
Upper Left Corner: 1,1
Description: GeoTIFF File Imported

Rectified /Reference Image

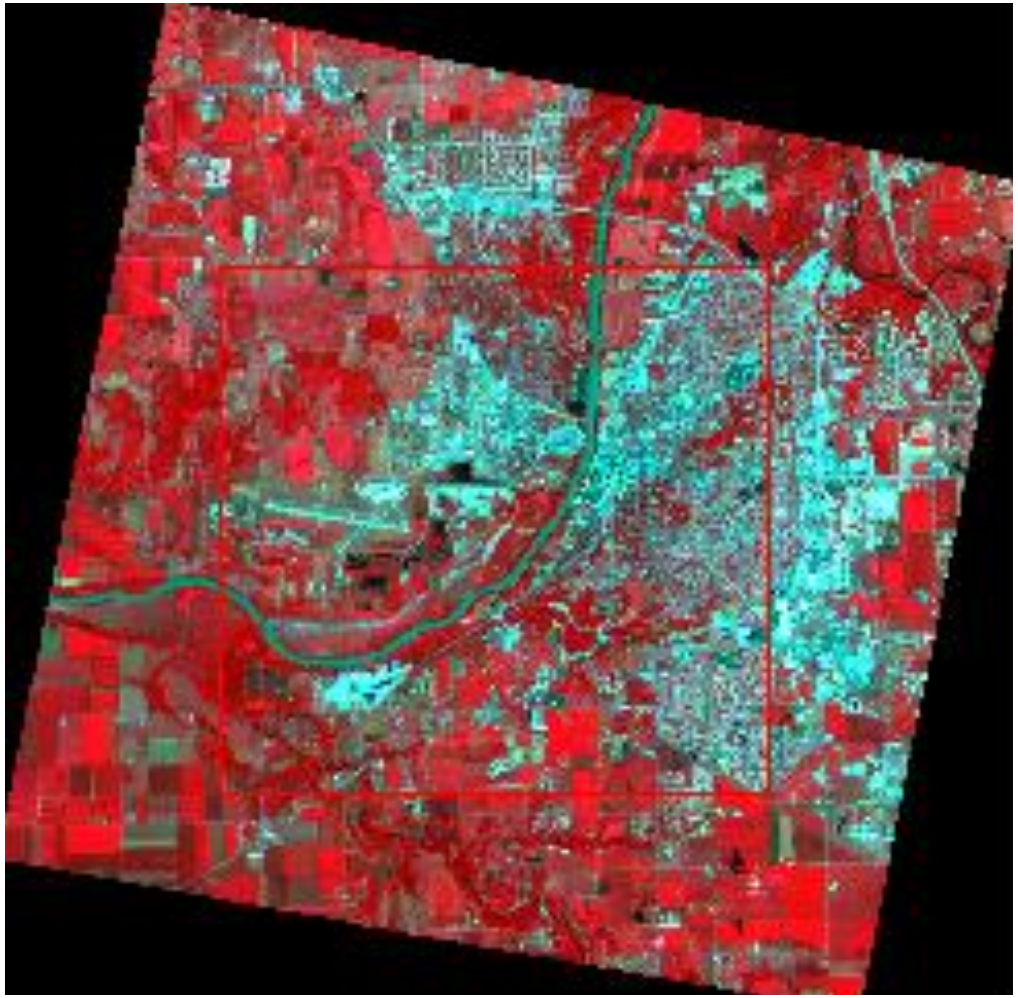


File Information:

File: F:\spotwarp & cl\spot _subset1
Dims: 626 x 614 x 3 [BIL]
Size: [Byte] 1,153,092 bytes.
File Type : ENVI Standard
Sensor Type: Unknown
Byte Order : Host (Intel)
Projection : None
Wavelength : None
Upper Left Corner: 2085,485
Description: File Resize Result, x resize factor: 1.000000, y resize factor: 1.000000. [Tue Aug 07 12:08:38 2001]

Un rectified Image

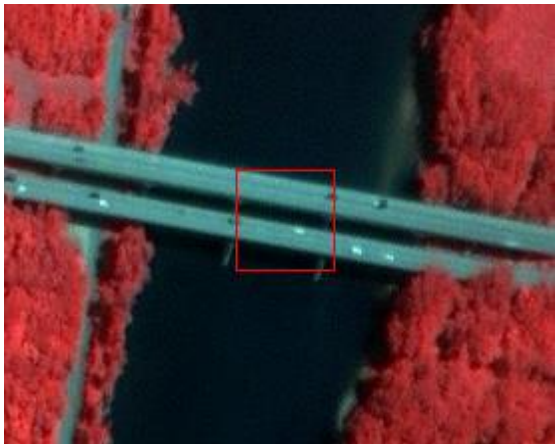
Georeferenced Image

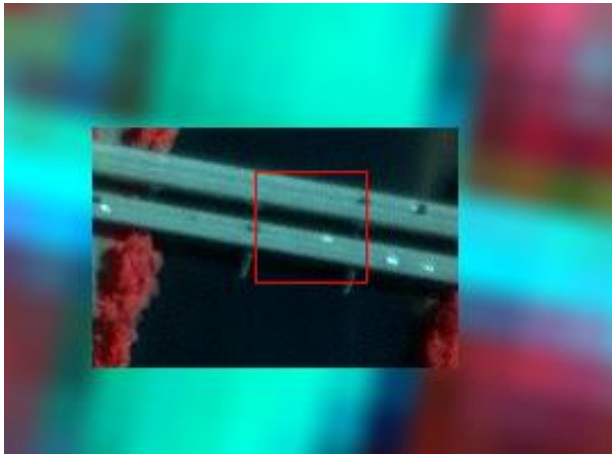


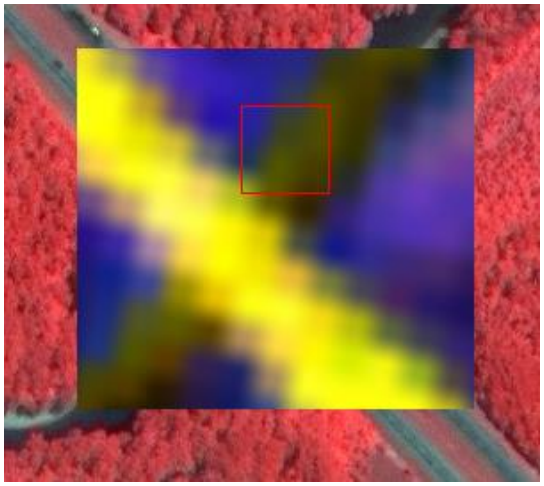
File Information:

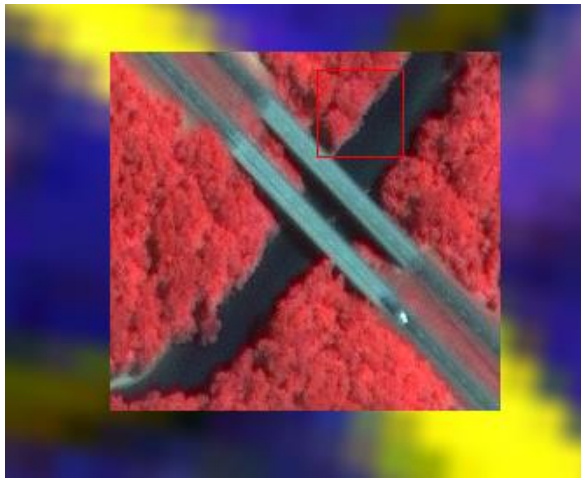
File: F:\spotwarp & cl\spot_ikreg20m
Dims: 732 x 722 x 3 [BIL]
Size: [Byte] 1,585,512 bytes.
File Type : ENVI Standard
Sensor Type: Unknown
Byte Order : Host (Intel)
Projection : UTM, Zone 16 North
Datum : North America 1983
X Pixel : 20.00000 Meters
Y Pixel : 20.00000 Meters
Wavelength : None
Upper Left Corner: 1,1
Description: Registration Result.

Overlaying After Geo-referencing





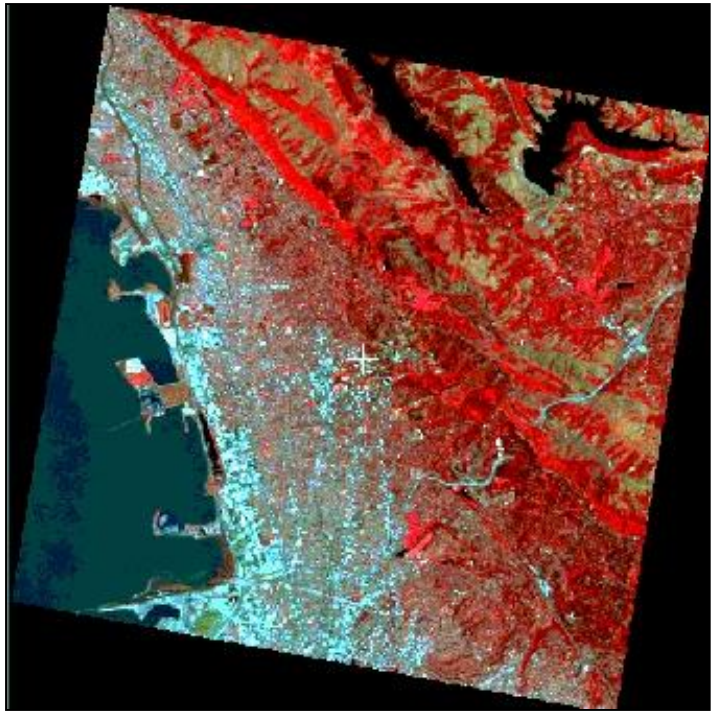




Geo-referencing



Before



After

Orthorectification



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